EPGSpeller: Open-Vocabulary Silent Spelling with Electropalatography

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Abstract

We study open-vocabulary silent spelling from electropalatography (EPG) under a word-holdout protocol and ask whether spatial inductive biases derived from a fixed electrode-to-grid proxy layout improve character-level decoding. We evaluate three EPG front-end representations under matched training conditions: (i) a layout-agnostic channel vector encoder, (ii) a layout-aware two-dimensional reconstruction consumed by convolution, and (iii) a layout-aware row/column compression. In this setting, the layout-aware two-dimensional convolutional front-end does not outperform the vector baseline, while spatial augmentation improves robustness to fixed electrode dropouts. All reported numbers are generated from pinned evidence files via a deterministic script.

1 Problem setting

Electropalatography (EPG) measures tongue–palate contact as a multichannel time series from a fixed electrode array. We work with a fixed inventory of 124electrode channels indexed by the SmartPalate channel numbering, and we use that index order as the canonical channel order for layout-agnostic models.

To introduce spatial structure, we define a proxy spatial map from channel indices to coordinates on a 16×16grid, derived from the pinned SmartPalate distribution artifact. A channel is *mapped* if it has an assigned grid coordinate (120), *unmapped* otherwise (4); 5mapped channels share a grid cell with at least one other channel and are aggregated deterministically when constructing a grid frame. This proxy mapping is fixed and used only as an inductive bias for constructing layout-aware representations; we do not claim it as a geometric tongue model.

We use a character-level CTC-style formulation [1]. In the word-holdout protocol, test words are unseen during training, so greedy CTC decoding is not constrained to a fixed lexicon.

2 Related work and positioning

Silent spelling has been explored in prior work (e.g., [2]). Our focus is on electropalatography and on isolating the effect of layout-agnostic versus layout-aware EPG front-end representations, while keeping the decoder and training protocol fixed.

3 Models and spatial front-ends

All variants share the same temporal backbone and CTC decoding setup. The temporal backbone produces 64-dimensional per-frame embeddings and uses the same recurrent stack across conditions (uni˙gru; 5layers; 512hidden units). We apply a temporal unfolding operator with kernel length 32and stride 4before the recurrent stack; we vary only the EPG spatial front-end described below.

Vector (layout-agnostic). The vector baseline uses the canonical channel index order and does not use any electrode layout information.

Layout-aware two-dimensional reconstruction. We scatter the selected channels into the 16×16proxy grid and apply a two-dimensional convolutional front-end to obtain per-frame embeddings. Grid cells without electrodes are masked, duplicated channels are aggregated deterministically, and unmapped channels are embedded separately and combined with the grid embedding.

Row/column compression. We form row-wise and column-wise pooled summaries from the same proxy grid, concatenate them (optionally with the unmapped channels), and project to the shared embedding size.

4 Experimental protocol

We evaluate the word-holdout protocol using 4random split seeds (seed0, seed1, seed2, seed3) and reuse the exact same splits across all compared spatial front-ends. Training uses 10000optimization steps at 10.0 ms frame rate and enables SpecAugment-inspired masking [3].

We compare two simple electrode-selection heuristics for budgeted subsets: TopK (importance-based ranking) and FPS-diverse TopK (importance ranking with an additional spatial-diversity constraint on the proxy grid). We treat both as baselines and do not claim novelty for these heuristics.

We report greedy CER (greedy CTC decoding) and streaming RTF (a streaming speed metric) throughout; lower CER and lower RTF are better.

5 Results

Scope. All comparisons below use the same protocol and matched training settings; we interpret the findings as evidence about spatial inductive biases under this setting.

5.1 Two-dimensional convolution vs vector

Table 1 compares vector and two-dimensional convolutional front-ends for multiple electrode budgets. In this protocol and setting, the layout-aware two-dimensional convolutional front-end is worse than the vector baseline in greedy CER and also slower in streaming RTF.

Table 1: Vector vs two-dimensional convolution (mean±std over splits).

K	Vector Greedy CER	Spatial2D Greedy CER	Vector Stream RTF	Spatial2D Stream RTF
TopK				
32	0.1399 ± 0.0162	0.4885 ± 0.1269	0.000491 ± 0.000035	0.002670 ± 0.000127
64	0.1289 ± 0.0135	0.2609 ± 0.0300	0.000476 ± 0.000004	0.002895 ± 0.000304
96	0.1194 ± 0.0173	0.2304 ± 0.0350	0.000475 ± 0.000023	0.002767 ± 0.000238
124	0.1048 ± 0.0093	0.1831 ± 0.0103	0.000469 ± 0.000014	0.002639 ± 0.000067
FPS2K				
32	0.1440 ± 0.0136	0.5795 ± 0.3052	0.000514 ± 0.000022	0.002674 ± 0.000129
64	0.1195 ± 0.0103	0.3581 ± 0.3079	0.000532 ± 0.000094	0.002808 ± 0.000180
96	0.1137 ± 0.0161	0.3438 ± 0.3104	0.000486 ± 0.000029	0.002626 ± 0.000096
124	0.1048 ± 0.0093	0.1831 ± 0.0103	0.000469 ± 0.000014	0.002639 ± 0.000067

Table 2: Vector vs row/column compression (mean±std over splits).

K	Vector Greedy CER	RowCol Greedy CER	Vector Stream RTF	RowCol Stream RTF
TopK				
32	0.1399 ± 0.0162	0.1448 ± 0.0204	0.000491 ± 0.000035	0.002141 ± 0.000276
64	0.1289 ± 0.0135	0.1269 ± 0.0143	0.000476 ± 0.000004	0.002072 ± 0.000142
FPS2K				
32	0.1440 ± 0.0136	0.1466 ± 0.0080	0.000514 ± 0.000022	0.002176 ± 0.000311
64	0.1195 ± 0.0103	0.1255 ± 0.0193	0.000532 ± 0.000094	0.002104 ± 0.000216

5.2 Row/column compression vs vector

Table 2 shows that row/column compression stays close to the vector baseline in greedy CER, but increases streaming RTF.

5.3 Spatial augmentation and robustness to electrode failures

To quantify robustness, we apply fixed electrode-dropout masks at evaluation time and report the increase in greedy CER relative to clean evaluation. Here, *spatial augmentation* refers to training-time masking or perturbations that operate on the proxy electrode grid (rather than treating channels as exchangeable). Table 3 shows that spatial augmentation reduces the CER degradation under dropout.

6 Artifacts and auditability

This project maintains a claim–evidence registry in `paper/paper.json`. All numeric facts and result tables in this draft are generated from pinned evidence snapshots via `scripts/paper/generate_facts_tex.py`

Table 3: Fixed-dropout robustness for the two-dimensional front-end (mean±std over splits). Values are ΔCER relative to clean evaluation.

Drop rate	No spatial aug	Spatial aug
0.0	0.0000 ± 0.0000	0.0000 ± 0.0000
0.1	0.0245 ± 0.0027	0.0183 ± 0.0065
0.2	0.0783 ± 0.0162	0.0660 ± 0.0097
0.3	0.1440 ± 0.0183	0.1224 ± 0.0120

using `paper/facts_spec.yaml`. A strict checker rejects handwritten numeric facts in the manuscript body.

Key pinned evidence identifiers used by this draft include `ev_scripts_smartpalate_distribution_csv` (`d21d93c9ec3c`), `ev_ed20260217_h11_spatial2d_vs_vector_p1_k32_k64_k96_k124_2026-02-17` (`2f2840f6dee5`), `ev_ed20260217_h13_rowcol_vs_vector_p1_k32_k64_2026-02-17` (`a7fb56ec370d`), and `ev_ed20260217_h12_spatial_aug_fixed_dropout_summary_p1_topk_k64_2026-02-17` (`0cd1fecedc16`).

References

[1] Alex Graves, Santiago Fernández, Faustino Gomez, and Jürgen Schmidhuber. Connectionist temporal classification, 2006.

[2] Naoki Kimura, Tan Gemicioglu, Jonathan Womack, Richard Li, Yuhui Zhao, Abdelkareem Bedri, Zixiong Su, Alex Olwal, Jun Rekimoto, and Thad Starner. Silentspeller: Towards mobile, hands-free, silent speech text entry using electropalatography, 2022.

[3] Daniel S. Park, William Chan, Yu Zhang, Chung-Cheng Chiu, Barret Zoph, Ekin D. Cubuk, and Quoc V. Le. Specaugment: A simple data augmentation method for automatic speech recognition, 2019.